KEDGE Business School EBP1 Quantitative Methods 1

Worksheet 1 — Presenting and organising data

Exercise statements

Badr ABOUFARASSE, PhD — 2026–2027

1 Core applications

Master the essential definitions and calculations from the course.

Core application — 10 min

1.1 Exercise 1 – Population, sample and variable types

A store wants to study all customers who made a purchase in September. It surveys a sample of 80 customers and records:

Variable	Example
Payment method	card, cash, mobile
Number of items purchased	1, 2, 3, ...
Time spent in store	minutes
Satisfaction level	very dissatisfied to very satisfied

1. Identify the target population, the sample and the statistical unit.
2. Classify each variable as nominal qualitative, ordinal qualitative, discrete quantitative or continuous quantitative.
3. Choose an appropriate graph for each variable.
4. For which variables do cumulative frequencies make sense? Explain.
5. Explain why a large sample is not necessarily representative.

Core application — 20 min

1.2 Exercise 2 – Discrete distribution

x_i	0	1	2	3	4	5
n_i	3	7	12	10	6	2

1. Check the total count N.
2. Calculate relative frequencies $f_i = n_i/N$ and increasing cumulative frequencies.
3. Draw a stick/bar diagram.
4. Construct the cumulative distribution function $F(x) = P(X \leq x)$.
5. Find the proportion of customers who placed at most three orders.

Core application — 15 min

1.3 Extra 1 — Cross-tabulation and conditional percentages

Among 120 students, 50 use public transport, including 30 women. Of the 70 other students, 35 are women.

1. Build the gender × transport table.
2. Compute the public-transport rate within each gender.

3. State whether an association appears present.

Method hint:

Conditional percentages use the total of the conditioning group as denominator.

2 Assessment and exam practice

Practise independently with assessment and final-exam style questions.

Assessment / exam practice — 40 min

2.1 Exercise 3 – Continuous variable with unequal class widths

Class	$[10, 14[$	$[14, 18[$	$[18, 26[$	$[26, 30[$	$[30, 35[$	$[35, 40[$	$[40, 45[$	$[45, 50[$	$[50, 55[$
n_i	4	12	32	32	60	35	25	5	5

1. For each class, calculate width a_i, relative frequency f_i and count density $d_i = n_i/a_i$.
2. Draw the correct histogram and explain why raw counts cannot be used as heights when class widths differ.
3. Calculate increasing and decreasing cumulative counts and the corresponding cumulative relative frequencies.
4. Draw the cumulative curves.
5. Find the number and proportion willing to pay less than 35 euros, then at least 35 euros.

Assessment / exam practice — 35 min

2.2 Extra 3 — Continuous-assessment case: willingness to pay

Case study — 12 marks. A food retailer asks 200 customers for the maximum price they would pay for a sustainable lunch box.

Maximum price (€)	[20,25[	[25,30[	[30,40[	[40,50[	[50,60[	[60,80[
Customers	10	20	50	60	40	20

1. Identify the population, statistical unit, variable and variable type. *(2 marks)*
2. Calculate class widths, relative frequencies and frequency densities. *(3 marks)*
3. Explain how to construct the correct histogram and identify the modal class. *(2 marks)*
4. Calculate ascending and descending cumulative counts. Give the coordinates needed to draw both cumulative curves. *(3 marks)*
5. Find the highest price that at least 60% of surveyed customers would be willing to pay. Explain the percentile used. *(2 marks)*

Method hint:

Unequal-width classes require frequency density. For the pricing decision, translate “at least 60% can pay” into an ascending cumulative percentage.

Assessment / exam practice — 30 min

2.3 Extra 4 — Exam-style synthesis: reporting customer satisfaction

Independent case — 10 marks. Two stores collect satisfaction ratings on an ordered scale from 1 (very dissatisfied) to 5 (very satisfied).

Rating	1	2	3	4	5	Total
Store A	6	10	18	36	30	100
Store B	4	12	30	34	20	100

1. Identify the variable and its measurement scale. *(1 mark)*
2. Choose and justify one graph for comparing the two stores. Describe precisely how it should be constructed. *(2 marks)*
3. Compute, for each store, the percentages rated 4 or 5 and the cumulative percentage rated at most 2. *(3 marks)*
4. A manager says: "Store A is 50% better because it has 30 ratings of 5 versus 20." Criticise this claim. *(2 marks)*
5. Write a three-sentence statistical conclusion, including one limitation of the survey. *(2 marks)*

Method hint:

Keep the order of the categories, compare percentages because the store totals could differ in another dataset, and separate description from causation.

3 Optional advanced practice

Explore additional interpretations and professional applications.

Advanced practice — 15 min

3.1 Extra 2 — Data-quality and communication protocol

A CRM extract contains duplicate customers, missing ages coded 999, inconsistent dates and revenue recorded in euros and thousands of euros. Propose a reproducible six-step protocol before producing a management dashboard.

Method hint:

Separate validation, cleaning, documentation and communication.